

```
import sys
```

```
def get_hematology_input():
```

```
    """Collect hematology lab results from user input."""
```

```
    print("\nEnter hematology results:")
```

```
    hematology = {
```

```
        'sex': input("Sex (male/female/pregnant): ").lower(),
```

```
        'Hb': float(input("Hemoglobin (g/dL): ")),
```

```
        'RBC': float(input("RBC count (million/ $\mu$ L): ")),
```

```
        'WBC': float(input("WBC count (cells/ $\mu$ L): ")),
```

```
        'MCV': float(input("MCV (fL): ")),
```

```
        'MCH': float(input("MCH (pg): ")),
```

```
        'MCHC': float(input("MCHC (g/dL): ")),
```

```
        'RDW': float(input("RDW (%): ")),
```

```
        'retic': float(input("Reticulocyte count (%): ")),
```

```
        'ferritin': float(input("Ferritin (ng/mL): ")),
```

```
        'TIBC': float(input("TIBC ( $\mu$ g/dL): ")),
```

```
        'SI': float(input("Serum Iron ( $\mu$ g/dL): ")),
```

```
        'LDH': float(input("LDH (U/L): ")),
```

```
        'total_bilirubin': float(input("Total Bilirubin (mg/dL): ")),
```

```
        'direct_bilirubin': float(input("Direct Bilirubin (mg/dL): ")),
```

```
        'AST': float(input("AST (U/L): ")),
```

```
        'coombs': input("Coombs Test (positive/negative): ").lower()
```

```
    }
```

```
    return hematology
```

```
def get_patient_history():
```

```
    """Collect patient history relevant to anemia diagnosis."""
```

```
    print("\nPatient History (yes/no):")
```

```
    history = {
```

```
        'pica': input("Pica (ice/dirt cravings)? ").lower() == 'yes',
```

```
        'vegetarian': input("Vegetarian diet? ").lower() == 'yes',
```

```
        'recent_travel': input("Recent travel to tropical areas? ").lower() == 'yes',
```

```
        'recent_infection': input("Recent infection (fever/URI symptoms)? ").lower() == 'yes',
```

```
        'sepsis': input("Signs of sepsis (SIRS)? ").lower() == 'yes',
```

```
        'fava_beans': input("Consumed fava beans (baghali)? ").lower() == 'yes',
```

```
        'menorrhagia': input("Heavy menstrual bleeding? ").lower() == 'yes',
```

```
        'GI_bleed': input("GI bleeding (melena/hematochezia)? ").lower() == 'yes',
```

```
        'dark_urine': input("Dark/cola-colored urine? ").lower() == 'yes',
```

```
        'B_symptoms': input("B symptoms (fever, night sweats, weight loss)? ").lower() == 'yes',
```

```
        'drugs': input("Taking MTX, azathioprine, hydroxyurea? ").lower() == 'yes',
```

```
        'family_anemia': input("Family history of anemia? ").lower() == 'yes',
```

```
        'opium': input("Opium use? ").lower() == 'yes',
```

```
        'alcohol': input("Heavy alcohol use? ").lower() == 'yes',
```

```
        'liver_disease': input("History of liver disease? ").lower() == 'yes',
```

```

        'neurologic': input("Neurologic symptoms (gait problems)? ").lower() == 'yes'
    }
    return history

def get_physical_exam():
    """Collect physical exam findings relevant to anemia."""
    print("\nPhysical Exam Findings (yes/no):")
    exam = {
        'shock': input("Signs of shock (hypotension, tachycardia)? ").lower() == 'yes',
        'cachexia': input("Cachexia/bitemporal wasting? ").lower() == 'yes',
        'icterus': input("Scleral icterus? ").lower() == 'yes',
        'palmar_erythema': input("Palmar erythema? ").lower() == 'yes',
        'LAD': input("Lymphadenopathy? ").lower() == 'yes',
        'goiter': input("Goiter? ").lower() == 'yes',
        'geographic_tongue': input("Geographic tongue? ").lower() == 'yes',
        'burton_line': input("Burton's line (lead line)? ").lower() == 'yes',
        'hepatosplenomegaly': input("Hepatosplenomegaly? ").lower() == 'yes',
        'koilonychia': input("Koilonychia (spoon nails)? ").lower() == 'yes',
        'petechiae': input("Petechiae/purpura? ").lower() == 'yes',
        'skin_icterus': input("Skin icterus? ").lower() == 'yes'
    }
    return exam

def calculate_rpi(hematology):
    """Calculate Reticulocyte Production Index."""
    if hematology['sex'] == 'male':
        return (hematology['retic'] * hematology['Hb']) / 15
    else:
        return (hematology['retic'] * hematology['Hb']) / 13

def determine_microcytic_anemia_cause(hematology, history, exam):
    """Determine cause of microcytic anemia (MCV < 80)."""
    causes = []
    iron_profile = {
        'SI_low': hematology['SI'] < 60,
        'TIBC_high': hematology['TIBC'] > 360,
        'ferritin_low': (hematology['ferritin'] < 30 if hematology['sex'] in ['female', 'pregnant']
                        else hematology['ferritin'] < 50)
    }

    # Iron deficiency anemia criteria
    if all(iron_profile.values()):
        causes.append("Iron Deficiency Anemia")
        if history['pica'] or history['menorrhagia'] or history['GI_bleed'] or exam['koilonychia']:
            causes[-1] += " (likely due to: "

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```

    if history['pica']:
        causes[-1] += "nutritional deficiency/pica, "
    if history['menorrhagia']:
        causes[-1] += "menorrhagia, "
    if history['GI_bleed']:
        causes[-1] += "GI bleeding, "
    if exam['koilonychia']:
        causes[-1] += "chronic deficiency"
    causes[-1] = causes[-1].rstrip(", ") + ")"

elif hematology['MCV']/hematology['RBC'] > 13:
    causes.append("Suggestive of Iron Deficiency Anemia (Mentzer index >13) rather than
thalassemia")

elif history['family_anemia']:
    causes.append("Consider Thalassemia (family history present)")

elif history['opium'] and exam['burton_line']:
    causes.append("Lead poisoning (opium use + Burton's line)")

return causes if causes else ["Consider other microcytic anemias (sideroblastic, chronic
disease)"]

def determine_macrocytic_anemia_cause(hematology, history, exam):
    """Determine cause of macrocytic anemia (MCV > 100)."""
    causes = []

    if history['vegetarian'] or history['alcohol'] or exam['geographic_tongue']:
        causes.append("Possible B12/Folate Deficiency")
        if history['vegetarian']:
            causes[-1] += " (dietary)"
        if history['alcohol']:
            causes[-1] += " (alcohol-related)"
        if exam['geographic_tongue']:
            causes[-1] += " (folate deficiency sign)"

    elif history['neurologic']:
        causes.append("B12 Deficiency with neurologic manifestations")

    elif history['drugs']:
        causes.append("Drug-induced macrocytosis (check medications)")

    elif history['liver_disease'] or exam['palmar_erythema']:
        causes.append("Liver disease-associated macrocytosis")

```

```
return causes if causes else ["Consider other macrocytic anemias (MDS, hypothyroidism)"]
```

```
def determine_normocytic_anemia_cause(hematology, history, exam):
```

```
    """Determine cause of normocytic anemia (MCV 80-100)."""
```

```
    causes = []
```

```
    if history['B_symptoms'] or exam['cachexia'] or exam['LAD'] or exam['hepatosplenomegaly']:
        causes.append("Consider hematologic malignancies (leukemia, lymphoma)")
```

```
    elif exam['icterus'] or exam['skin_icterus'] or history['dark_urine']:
```

```
        causes.append("Hemolytic anemia")
```

```
        if hematology['coombs'] == 'positive':
```

```
            causes[-1] += " (autoimmune)"
```

```
        elif history['fava_beans']:
```

```
            causes[-1] += " (G6PD deficiency suspected)"
```

```
        else:
```

```
            causes[-1] += " (workup needed for exact cause)"
```

```
    elif history['GI_bleed']:
```

```
        causes.append("Acute blood loss anemia")
```

```
    elif history['recent_infection'] or history['sepsis']:
```

```
        causes.append("Anemia of inflammation/infection")
```

```
    return causes if causes else ["Consider anemia of chronic disease, renal disease, or early iron deficiency"]
```

```
def evaluate_anemia():
```

```
    """Main function to evaluate anemia based on inputs."""
```

```
    print("\n=== Anemia Evaluation Assistant ===")
```

```
    hematology = get_hematology_input()
```

```
    history = get_patient_history()
```

```
    exam = get_physical_exam()
```

```
    # Set hemoglobin threshold based on sex
```

```
    if hematology['sex'] == 'male':
```

```
        hb_threshold = 13
```

```
    elif hematology['sex'] == 'pregnant':
```

```
        hb_threshold = 10
```

```
    else:
```

```
        hb_threshold = 12
```

```
    rpi = calculate_rpi(hematology)
```

```

print("\n=== Evaluation Results ===")
print(f"Anemia present (Hb < {hb_threshold} g/dL): {'Yes' if hematology['Hb'] <
hb_threshold else 'No'}")
print(f"RPI: {rpi:.2f} ({'Appropriate' if rpi > 2.5 else 'Inappropriate'} response)")
print(f"MCV Classification: ", end="")

if hematology['MCV'] < 80:
    print("Microcytic")
    causes = determine_microcytic_anemia_cause(hematology, history, exam)
elif hematology['MCV'] > 100:
    print("Macrocytic")
    causes = determine_macrocytic_anemia_cause(hematology, history, exam)
else:
    print("Normocytic")
    causes = determine_normocytic_anemia_cause(hematology, history, exam)

print("\nPotential Causes:")
for i, cause in enumerate(causes, 1):
    print(f"{i}. {cause}")

print("\nRecommended Workup:")
if hematology['MCV'] < 80 and not hematology['TIBC']:
    print("- Complete iron profile (SI, TIBC, ferritin)")
    print("- Consider Hb electrophoresis if iron studies normal")
elif hematology['MCV'] > 100:
    print("- B12 and folate levels")
    print("- TSH if hypothyroidism suspected")
else:
    print("- Further evaluation based on suspected cause:")
    print(" - CBC with differential, peripheral smear")
    print(" - Renal function tests, LFTs")
    print(" - Inflammatory markers if chronic disease suspected")

if rpi > 2.5:
    print("- Evaluate for acute blood loss or hemolysis")

print("\n=== Medical Disclaimer ===")
print("This tool is for educational purposes only and does not replace clinical judgment.")
print("Always consult with a qualified physician for medical decisions.")

if __name__ == "__main__":
    evaluate_anemia()

```